La Scuola d'Italia / Dr. Huson / IB Math 11
7 October 2026

Nickname: SOLUTIONS

1.10 Quiz: Arithmetic and Geometric Sequences

A calculator is allowed. Show your working: write the formula you use, then substitute.

1. [Maximum mark: 8]

   Consider the arithmetic sequence

   $$15,\ 23,\ 31,\ 39,\ \ldots$$

   (a) Write down the value of the common difference d. [1]

   (b) Find an expression for u_n, the nth term. [2]

   (c) Find u_{30}. [2]

   (d) Show that 303 is a term of the sequence, and find which term it is. [3]

(a) $d = 8$

(b) $u_n = 15 + 8(n-1)$

(c) $u_{30} = 15 + 8(30-1)$
$= 247$

(d) $u_n = 15 + 8(n-1) = 303$
$8(n-1) = 288$
$n-1 = 36$
$n = 37$
$u_{37} = 303$ $\quad 37 \in \mathbb{N}$

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2. [Maximum mark: 9]

In an arithmetic sequence, $u_5 = 38$ and $u_{14} = 11$.

(a) Find the common difference d and the first term u_1. [3]

(b) Find the first term of the sequence that is negative. [3]

The first three terms of a different arithmetic sequence are

$$2k + 1, \qquad 4k - 3, \qquad 5k + 1.$$

(c) Find the value of k, and write down the first three terms. [3]

(a) $u_{14} - u_5 = [u_1 + d(14-1)] - [u_1 + d(5-1)]$

$= 11 - 38$

$9d = -27$

$d = -3$

$u_5 = u_1 + (-3)(5-1) = 38$

$u_1 = 50$

(b) $u_n = 50 - 3(n-1) < 0$

$n - 1 > \frac{50}{3} = 16\frac{2}{3}$

1st whole n: $n = 18$

$u_{18} = 50 - 3(18-1) = -1$

(c) $(4k-3) - (2k+1) = (5k+1) - (4k-3)$

$2k - 4 = k + 4$

$k = 8$

17, 29, 41

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3. [Maximum mark: 10]

(a) State whether each sequence is arithmetic, geometric or neither. Write down d or r where there is one. [4]

(i) 5, 10, 20, 40 (ii) 7, 4, 1, −2 (iii) 2, 3, 5, 8 (iv) 162, 54, 18, 6

Consider the geometric sequence

16, 20, 25, ...

(b) Find the common ratio r. [1]

(c) Find u_{12}. [2]

(d) Challenge: A different geometric sequence has $u_3 = 45$ and $u_6 = 1215$. Find the common ratio and the first term. [3]

(a) i) geometric $r = 2$
ii) arithmetic $d = -3$
iii) neither
iv) geometric $r = \frac{1}{3}$

(b) $r = \frac{20}{16} = 1.25$

(c) $u_{12} = 16 \cdot (1.25)^{12-1} = 186.264\ldots$
≈ 186

(d) $\frac{u_6}{u_3} = \frac{u_1 \cdot r^{6-1}}{u_1 \cdot r^{3-1}} = r^3 = \frac{1215}{45}$
$r^3 = 27$
$r = 3$
$u_3 = u_1 3^{3-1} = 45$
$u_1 = 5$

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4. [Maximum mark: 7]

Marco is offered two jobs.

Job A: \$38 000 in the first year, rising by \$2000 each year.
Job B: \$36 000 in the first year, rising by 6% each year.

(a) Write down an expression for the salary in year n for each job. [2]

(b) Find Marco's salary in year 3 for each job. [2]

(c) Find the first year in which the Job B salary is greater than the Job A salary. [3]

(a) $A_n = 38{,}000 + 2000(n-1)$

$B_n = 36{,}000 \cdot 1.06^{(n-1)}$

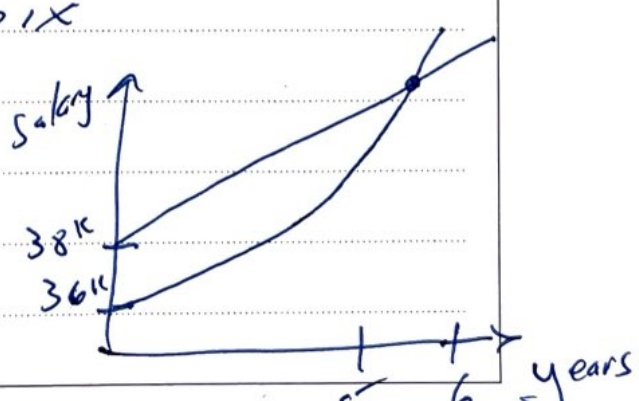
(b) $A_3 = 38{,}000 + 2000(3-1) = 42{,}000$

$B_3 = 36{,}000 \times 1.06^{3-1} = 40{,}449.60$

(c)

year	A	B
5	46,000	45,449.17
6	48,000	48,176.12 ←

year Six



sketch of graph